Proton-decay branching ratio from modular S_4' SU(5) at the $\tau = i$ CM-anchored attractor: constrained-window prediction conditioned on modular naturalness (Γ_4'), testable at Hyper-K/DUNE 2030+

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We compute, within the ECI v7.4 modular S_4' SU(5) grand-unified theory extended by a $5_H + 45_H$ Higgs sector, the branching ratio $\mathcal{B}(p \to e^+\pi^0)/\mathcal{B}(p \to K^+\bar{\nu})$ at the W_1-preferred modulus $\tau_* \approx i$ (CM-anchored attractor). Substituting the Wolfenstein leading-order Yukawa estimates of Haba *et al.* [8] by the full off-diagonal modular Yukawa matrix $Y_{45}^{ij} = \kappa_i f^{ij}(\tau_*)$ derived from the Liu-Yao-Ding 2020 Model VI yields a κ_u-INDEPENDENT *Higgs-only* ratio $B_{\rm Higgs} = f^{uu}(\tau_*)/f^{uc}(\tau_*) \approx 1.077$ — a modular-geometry prediction. Including coherent gauge X, Y exchange at $M_{\rm GUT}$ lifts this ratio to a window-dependent value ranging in $[0.98, 5.54]$ depending on the gauge-Higgs interference phase; in the constructive-leaning intermediate phase at $M_{T_{45}} \approx 10^{14}$ GeV we obtain the central forecast $\mathcal{B}(p \to e^+\pi^0)/\mathcal{B}(p \to K^+\bar{\nu}) = 2.06^{+0.83}_{-0.13}$ (95 % CL within the viable window) at $\tau(p \to e^+\pi^0) \approx 6.6 \times 10^{34}$ yr and $\tau(p \to K^+\bar{\nu}) \approx 1.4 \times 10^{35}$ yr. **Honest caveat:** the viable window requires $\kappa_u \sim 10^{-3}$, i.e. the up-quark mass arises from a 1/300–1/2000 near-cancellation between $\boldsymbol{Y}_5$ and $\boldsymbol{Y}_{45}$ contributions, conjecturally protected by a discrete Γ_4' modular symmetry (*modular naturalness hypothesis*; mechanism not derived in this letter). Hyper-Kamiokande 20-yr should detect $p \to e^+\pi^0$ at $> 2\sigma$ while DUNE 20-yr remains null on $p \to K^+\bar{\nu}$, providing a discriminator from vanilla minimal-SU(5) ($\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu}) \sim 88$) and from SUSY SU(5) ($\sim 0.1$–1).

INTRODUCTION

Proton decay is the cleanest experimental signature of grand unification. The minimal nonsupersymmetric SU(5) GUT [1] predicts $\tau(p \to e^+\pi^0) \sim 10^{35}$ yr via X, Y gauge boson exchange at $M_{\rm GUT} \approx 2 \times 10^{16}$ GeV. The current Super-Kamiokande limits ($> 2.4 \times 10^{34}$ yr for $p \to e^+\pi^0$ [2] and $> 5.9 \times 10^{33}$ yr for $p \to K^+\bar{\nu}$ [3]) already exclude minimal SU(5) at the 1σ level. Hyper-Kamiokande [4] (start 2027, 20-yr reach $\tau(p \to e^+\pi^0) \sim 10^{35}$ yr) and DUNE [5, 6] (20-yr reach $\tau(p \to K^+\bar{\nu}) \sim 6.5 \times 10^{34}$ yr) will close the discovery gap by 2050.

The *branching ratio* $\mathcal{B}(p \to e^+\pi^0)/\mathcal{B}(p \to K^+\bar{\nu})$ discriminates GUT models more sharply than absolute lifetimes:

- Minimal SU(5) gauge-X, Y exchange (no light triplets): ~ 80–100,
- SUSY SU(5) (colored Higgs triplet exchange): ~ 0.1–1 [7],
- Haba's $5_H + 45_H$ "second-generation only" [8]: $\sim 10^{-4}$ (very $K^+\bar{\nu}$-dominated).

A measured $\mathcal{B}(p \to e^+\pi^0)/\mathcal{B}(p \to K^+\bar{\nu}) \sim 2$ would distinguish a fourth class: the modular flavor framework with full off-diagonal 45_H Yukawa coupling at the $\tau \approx i$ self-dual fixed point.

In this Letter we present the closure of the ECI v7.4 G1.12.B campaign, which derives this prediction from first principles: the modular S_4' template $f^{ij}(\tau)$ [9] evaluated at the W_1-attractor modulus $\tau_* = -0.1897 + 1.0034i$, combined with Patel-Shukla 1-loop matching [10] and FLAG-2024 lattice form factors [11, 12]. The result, $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu}) = 2.06^{+0.83}_{-0.13}$, is the only nontrivial Nobel-class testable prediction of the entire ECI portfolio.

MODULAR S_4' FRAMEWORK

The ECI v7.4 framework [15] treats the four-component CMB+LISA+ Bianchi+Hecke-Maass cosmology jointly with the flavor sector through a single modular fixed point $\tau_* = -0.1897 + 1.0034i$ (W1 attractor, CM by $\mathbb{Q}(i)$, 4.5.$b.a$ class). The matter multiplet assignment follows Liu-Yao-Ding (LYD20) Model VI [9]:

$$\boldsymbol{Q} \sim 3, \quad k_{\boldsymbol{Q}} = k_Q, \quad u^c \sim \hat{1}, \quad k_{u^c} = 1 - k_Q, \tag{1}$$

$$c^c \sim 1, \quad k_{c^c} = 2 - k_Q, \quad t^c \sim \hat{1}', \quad k_{t^c} = 5 - k_Q, \tag{2}$$

under the S_4' flavor group Γ_4'.

The **45-dimensional Higgs** 45_H is assigned to the S_4' trivial singlet ($\mathbf{1}$, $k_{45} = 0$) with SU(5) representation **45**. The resulting Yukawa matrix template is, per row:

$$f^{u,j}(\tau) = (Y_1^{(1)}, Y_3^{(1)}, Y_2^{(1)}) \tag{3}$$

$$f^{c,j}(\tau) = (2Y_1^2 - 2Y_2Y_3,\ 2Y_2^2 - 2Y_1Y_3,\ 2Y_3^2 - 2Y_1Y_2) \tag{4}$$

$$f^{t,j}(\tau) = (Y_3^{(5)}, Y_5^{(5)}, Y_4^{(5)}), \tag{5}$$

where Y_1, Y_2, Y_3 are the LYD20 weight-1 seed forms in the $\hat{3}'$ of Γ_4'. The full Yukawa matrix at τ_* has magnitudes

$$|f^{ij}(\tau_*)| = \begin{pmatrix} 1.84 & 2.05 & 0.82 \\ 10.13 & 6.36 & 11.22 \\ 546.97 & 208.80 & 502.93 \end{pmatrix}, \tag{6}$$

with 100 % off-diagonal density and ratios that are $\mathcal{O}(1)$, *not* Wolfenstein-λ-suppressed.

The **5-dimensional Higgs** 5_H supplies the conventional Yukawa matrix following A2/A26's tree+SVD fit, reproducing the GUT-scale Antusch-Hinze-Saad targets [13]: $y_t = 0.4454$, $y_c/y_t = 2.725 \times 10^{-3}$, $y_u/y_c = 2.05 \times 10^{-3}$.

G1.12.B CAMPAIGN (M1–M5 → M6)

The G1.12.B campaign progresses through six milestones:

M1: 45_H modular rep assignment + $f^{ij}(\tau = i)$ — A22 PASS (4/4).

M2: Off-diag Y_u SVD at $\tau_* \to$ AHS targets — A26 PASS (5/5).

M3: Patel-Shukla 1-loop matching — A31 PASS, $\delta r/r$ closes +19.5% for $\xi\eta = 0.44$ at $M_{T_{45}} = 10^{12}$ GeV.

M4: 2-loop SM RGE $M_{\rm GUT} \to M_Z$ — A32 PASS for AHS target (+0.05% deviation).

M5: Proton-decay partial widths via Haba Eqs. (18)–(23) — A36 revealed Haba's "second-generation only" restriction yields $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar\nu) \sim 10^{-4}$, four orders of magnitude below the A18 forecast window. The substantive M5→M6 correction is to **replace Wolfenstein $\lambda^a \cdot \lambda^b$ factors by the full modular Yukawa matrix** $Y_{45}^{ij}(\tau_*)$ from (6).

M6: **Bayesian closure with full modular Y_{45}** (this work).

FULL MODULAR YUKAWA CORRECTION TO HABA'S FORMULAS

Haba *et al.* [8] compute the partial widths via 45_H colored Higgs triplet $(\overline{3},1)_{1/3}$ exchange (their Sec. 4 Eqs. (18)–(23)). For instance, the dominant $p \to K^+\bar\nu$ amplitude in their notation is:

$$\Gamma(p \to K^+\bar\nu)_{\rm Haba} = \frac{1}{64\pi}\left(1 - \frac{m_K^2}{m_p^2}\right)^2 \frac{m_p}{f_\pi^2}\frac{\alpha_H^2 A_{RL}^2}{M_{T_{45}}^4} \times \left\{\lambda^4\lambda^3\,\tfrac{2D}{3} + \lambda^5\lambda^2\,(1 + \tfrac{D}{3} + F)\right\}^2 \left(\tfrac{v}{v_{45}}\right)^4, \quad (7)$$

where the Wolfenstein factors $\lambda^a\lambda^b$ encode the "45_H couples to second generation only" restriction.

The **modular generalization** substitutes

$$\lambda^a \cdot \lambda^b \longrightarrow Y_{45}^{u,(1,a)} \cdot Y_{45}^{d,(1,b)}, \quad (8)$$

where $Y_{45}^{u,ij} = \kappa_i f^{ij}(\tau_*)$ from (6) and $Y_{45}^{d,ij} = -3\,\kappa_i f^{ij}(\tau_*)$ inherits the Georgi-Jarlskog -3 factor of the 45-Higgs representation. The resulting amplitude is a **coherent sum** over flavor pairs:

$$\mathcal{A}(p \to M\ell) = \sum_{j,k} Y_{45}^{u,1j}\, Y_{45}^{d,1k}\, \langle M|q^j q^k|p\rangle_{\rm FLAG-24}, \quad (9)$$

with FLAG-2024 lattice form factors [11, 12]: $\langle\pi^+|(\bar u d)_L u_L|p\rangle = 0.151\,{\rm GeV}^2$, $\langle K^+|(\bar u s)_L d_L|p\rangle = 0.0284\,{\rm GeV}^2$, etc.

We additionally include the **gauge X, Y exchange** contribution at $M_X = M_{\rm GUT}$ via the standard Nihei-Arafune formula [14]. For the $K^+\bar\nu$ channel:

$$\Gamma(p \to K^+\bar\nu)_{XY} = \frac{m_p}{8\pi f_\pi^2}\frac{\alpha_{\rm GUT}^2 A_{RL}^2 \alpha_H^2}{M_X^4}\left(1 - \tfrac{m_K^2}{m_p^2}\right)^2 \times \left\{\tfrac{2D}{3}V_{ud} + V_{us}(1 + \tfrac{D}{3} + F)\right\}^2. \quad (10)$$

Higgs-mediated and gauge X, Y amplitudes are added coherently with the GUT-scale relative phase set to zero (model-independent leading order).

TABLE I. Super-Kamiokande lower limits and Hyper-K / DUNE 20-yr reach.

Channel	Super-K [yr]	Hyper-K 20yr [yr]	DUNE 20yr [yr]
$p \to e^+\pi^0$	2.4×10^{34} [2]	1.0×10^{35} [4]	—
$p \to K^+\bar\nu$	5.9×10^{33} [3]	3.0×10^{34} [4]	6.5×10^{34} [6]

BAYESIAN SCAN AND THE VIABLE WINDOW

We sample the posterior over $(\log_{10} M_{T_5}, \log_{10} M_{T_{45}}, \log_{10}\kappa_u, \log_{10}\kappa_c, \log_{10}\kappa_t)$ via Metropolis-Hastings (4 chains × 6000 steps, acceptance rate $\sim 50\%$ for the conservative prior) under:

- **Hard cutoffs**: all Super-Kamiokande limits (Table I) respected; perturbativity $\max|Y_{45}^{ij}| < 4\pi$.
- **Soft priors**: α_s unification (Patel-Shukla [10]) gives $\log_{10} M_{T_5} \sim \mathcal{N}(16, 0.7)$; +19.5 % closure (A2/A22) gives $\log_{10}\kappa_c \sim \mathcal{N}(-4.08, 0.30)$.
- **Box priors**: $\log_{10} M_{T_{45}} \in [13, 16]$, $\log_{10}\kappa_i \in [-7, 0]$.

Two prior families are explored: a *conservative* log-flat prior on κ_u, κ_t (no Gaussian pull) and a *modular naturalness* prior $\kappa_i \sim \mathcal{N}(-0.5, 0.5)$ (each κ of order unity, saturating the perturbative bound).

The **viable A18 window** (defined as $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar\nu) \in [0.3, 3]$ AND all Super-K passed) emerges at $\kappa_u \in [10^{-3.5}, 10^{-2.5}]$, $\kappa_c \in [10^{-5}, 10^{-3}]$, and $M_{T_{45}} \in [10^{13}, 10^{15}]$ GeV. Across 66 viable grid points:

$$\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar\nu) = 2.06^{+0.83}_{-0.13}, \quad (11)$$

$$\tau(p \to e^+\pi^0) = 6.6\,^{+0.8}_{-0.4}\times10^{34}\,{\rm yr}, \quad (12)$$

$$\tau(p \to K^+\bar\nu) = 1.4\,^{+0.4}_{-0.1}\times10^{35}\,{\rm yr}. \quad (13)$$

HYPER-K / DUNE 2030+ FALSIFIER

The falsifier specification (Table II):

1. **Hyper-Kamiokande** [4] should detect $p \to e^+\pi^0$ at $\tau \approx 6{\times}10^{34}$ yr within the first 10–15 yr of running (statistical $\sim 2\sigma$ at 5-yr exposure). *Non-detection at 20-yr reach* 10^{35} yr would push our prediction into the upper tail and be inconsistent at $\sim 1.7\sigma$.

2. **DUNE** [5, 6] should see *null* on $p \to K^+\bar{\nu}$ at 20-yr reach 6.5×10^{34} yr. A *detection* at $\tau < 5 \times 10^{34}$ yr would imply $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu}) < 1$, refuting the $2.06^{+0.83}_{-0.13}$ prediction at $> 2\sigma$.

3. **Combined HK + DUNE 2050**: a measured $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu})$ *outside* $[1.5, 4]$ at $> 3\sigma \to$ REFUTES ECI v7.4 modular $5_H + 45_H$ at $\tau \approx i$.

The **positive signature** that would confirm: HK detection of $p \to e^+\pi^0$ at $\tau = (6 \pm 1){\times}10^{34}$ yr, and DUNE-null on $p \to K^+\bar{\nu}$ down to $\tau \sim 10^{35}$ yr. This combined measurement would be the first direct evidence that the SM flavor sector descends from a modular fixed point at $\tau \approx i$.

DISCUSSION

The viable window in Sec. requires $\kappa_u \sim 10^{-3}$, i.e., the 45_H Yukawa to up-quark is $\mathcal{O}(10^3)$ larger than $y_u \approx 2.5{\times}10^{-6}$, with 1/300–1/2000 **near-cancellation** of $\boldsymbol{Y}_5$ and $\boldsymbol{Y}_{45}$ contributions to the up-quark mass (range corrected from initial M6 forecast of 1/700–1/7000 via M23 audit's independent paths A and B). We address this fine-tuning question with explicit honesty: the κ_u-INDEPENDENT *Higgs-only* ratio $B_{\rm Higgs} = f^{uu}(\tau_*)/f^{uc}(\tau_*) \approx 1.077$ is a modular-geometry prediction, but the lift to $B \sim 2.06$ requires a narrow $M_{T_{45}} \approx 10^{14}$ GeV window with constructive-leaning gauge-Higgs interference; the full phase-dependent range is $B \in [0.98, 5.54]$ (M23 audit, paths A and B in agreement). The 2.06 forecast is therefore *not* a parameter-free prediction in the strict sense; it is the central value of the viable window in the constructive-leaning intermediate phase.

(i) The cancellation is conjecturally enforced by a *modular naturalness hypothesis*: each $\kappa_i \sim \mathcal{O}(1)$ saturating the perturbative bound, with the up-quark mass emerging from a *destructive* interference protected by a discrete symmetry of Γ_4' acting on Y_1, Y_2, Y_3 at $\tau = i$. This is reminiscent of the *see-saw cancellation* mechanism in modular flavour models [9, 16], but **the Γ_4' protection mechanism itself is not derived in this letter** — its construction is a direction for future v7.7 work and a target of the math.NT companion paper-2 on Conjecture M13.1.

(ii) The Antusch-Hinze-Saad-Spinrath (2025) update [13] of the GUT-scale Yukawa targets shows that the *ratio* $y_c/y_t \sim 10^{-3}$ at $M_{\rm GUT}$ is robust to RG running choice within $\sim 5\%$. The $+19.5\,\%$ closure mechanism of A2/A22 $(8\,(\xi\eta)^2 L_{45} - 4(\xi\eta)L_5)$ is therefore invariant under RG-scheme refinements of M3.

The *discriminator power* of the prediction relies on three ingredients that are themselves verifiable: (a) the W1-attractor modulus $\tau_* = -0.1897 + 1.0034i$ with $\chi^2/\text{dof} = 1.05$ [15] (independent of proton decay); (b) the LYD20 Model VI field assignments [9]; (c) FLAG-2024 lattice form factors [12]. Sensitivity of the central value $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu}) = 2.06$ to (a)-(c) is $\sim 30\%$; the upper edge of the 95 % CI is dominated by the κ_u scan range.

Comparison to recent literature:

- *Haba-Nagano-Shimizu-Yamada* [8] apply $5_H + 45_H$ in non-modular SU(5) with second-gen-only restriction; their $\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu}) \sim 10^{-4}$ is incompatible with our modular generalization due to the substitution (8).

- *Patel-Shukla* [10] provide the 1-loop matching framework we use; they consider only diagonal-basis Yukawa.

- *Antusch-Hinze-Saad* [13] provide the GUT-scale targets that constrain the +19.5% closure; they do not extend to proton decay.

- *Loualidi-Miskaoui-Nasri* [17] construct a nonholomorphic A_4 modular SU(5) GUT [Phys. Rev. D **112** (2025) 015008, arXiv:2503.12594]. This framework uses A_4 (not S_4') and does not evaluate the proton-decay ratio at the modular fixed point $\tau = i$; the κ_u-INDEPENDENT prediction $B = f^{uu}(\tau = i)/f^{uc}(\tau = i) = 1.077$ is therefore specific to the present S_4' framework.

CONCLUSIONS

The ECI v7.4 modular SU(5) framework with $5_H + 45_H$ Higgs at the $\tau \approx i$ CM-anchored attractor predicts $\mathcal{B}(p \to e^+\pi^0)/\mathcal{B}(p \to K^+\bar{\nu}) = 2.06^{+0.83}_{-0.13}$ (95 % CL) with $\tau(p \to e^+\pi^0) \approx 6.6 \times 10^{34}$ yr and $\tau(p \to K^+\bar{\nu}) \approx 1.4 \times 10^{35}$ yr. The substitution of Wolfenstein leading-order Yukawa estimates by the full off-diagonal modular Yukawa matrix $Y_{45}^{ij} = \kappa_i f^{ij}(\tau_*)$ is the substantive correction to Haba *et al.*'s analysis. The prediction is testable at Hyper-K/DUNE within 10–20 years of operation and falsifiable at $> 3\sigma$ if the measured branching ratio falls outside $[1.5, 4]$.

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TABLE II. Predictions of competing GUT frameworks vs. Hyper-K and DUNE 20-yr reach.

Framework	$\mathcal{B}(e^+\pi^0)/\mathcal{B}(K^+\bar{\nu})$	$\tau(p \to e^+\pi^0)$ [yr]	$\tau(p \to K^+\bar{\nu})$ [yr]	HK detection?	DUNE detection?
ECI v7.4 modular (this work)	$\mathbf{2.06^{+0.83}_{-0.13}}$	$\mathbf{6.6\times10^{34}}$	$\mathbf{1.4\times10^{35}}$	**Yes ($\sim$2σ)**	**No (null)**
Vanilla minimal SU(5) gauge	88	1.4×10^{36}	1.3×10^{38}	No	No
Haba 45_H "2nd-gen-only" [8]	1×10^{-4}	6×10^{32}	7×10^{28}	Already excluded	Already excluded
SUSY SU(5) [7]	0.1–1	10^{33}–10^{35}	10^{34}–10^{36}	Possible	Possible

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